

Lead

82 Pb Like tin, lead and its compounds have been used for about 5,000 years. This gray metal is soft, malleable, and resistant to rust. In the past, it was used to make many things, including coins, plumbing pipes, and paint glazes. Awareness about lead's toxicity has limited its applications today, which still include car batteries, electricity cables, and decorative stained glass.



Dull gray metal

Pure lead refined in a laboratory



Shiny, green glaze

Lead glaze

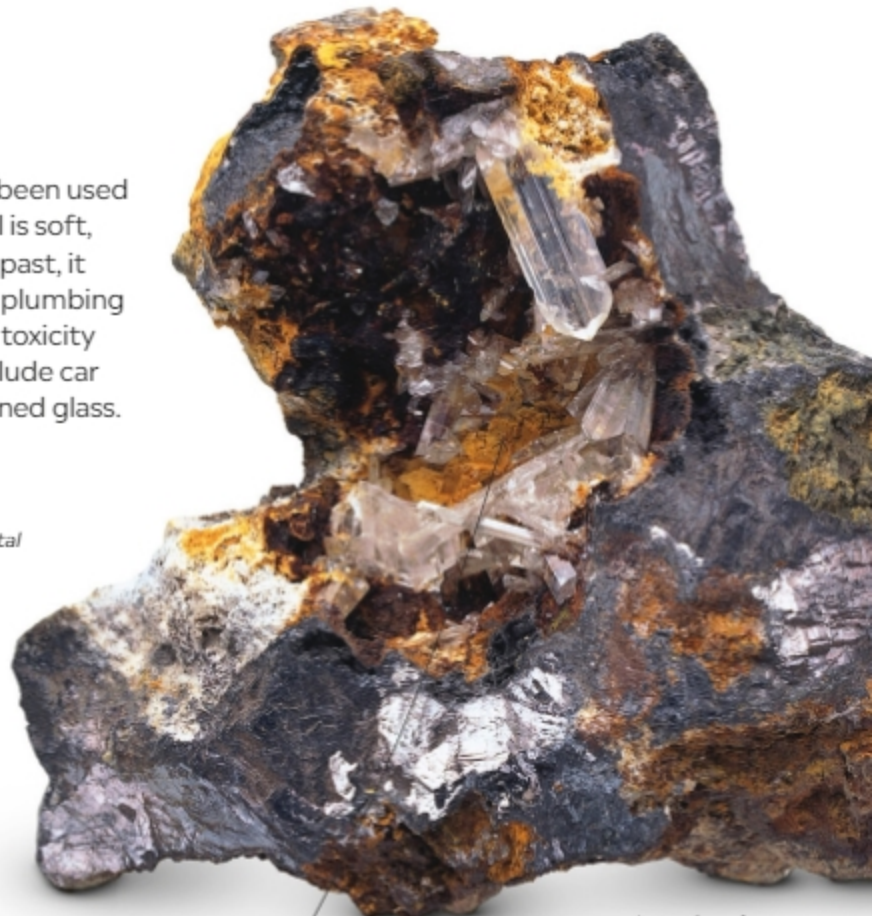
The green glaze on this model of a pigsty from ancient China is made of lead compounds. When such a coating is applied to pottery, it makes it waterproof. Once fired in a kiln, the lead glaze becomes shiny.

Lead power

The rechargeable batteries in gasoline and diesel cars are known as lead-acid batteries. This is because they contain lead plates, one of which is coated with lead oxide, in a solution of sulfuric acid. More modern electric vehicles are powered by lithium-ion batteries, but they sometimes have a lead-acid battery as well.



Modern lead-acid car batteries are sealed and require no maintenance.



Brownish-yellow crystals containing a lead compound

Chinese earthenware model of a pigsty from Later Han Dynasty, 206 BCE–220 CE

Anglesite

This mineral contains a lead-sulfur compound that is one of the main sources of lead. Anglesite is produced in nature when oxygen reacts with galena—another ore of lead.

Extra sparkle

Lead crystal—a type of decorative glassware—is glass that contains at least 24 percent lead oxide, which makes it heavier. It also bends light more than regular crystal and sparkles more as a result. Lead was used extensively in glassware before potential health risks over its use limited the manufacture of lead crystal.



Lead crystal has a distinctive gray shade and a bright finish.